hyper-thermochem: JAX-compatible and differentiable local kernels for hypersonic nonequilibrium CFD

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Local Closure Layer

Transitional hypersonic flows require closures that remain usable under strong thermochemical nonequilibrium, stiff source terms, and changing collisional regimes. `hyper-thermochem-py` is a reusable local property and source-term library for that setting: thermodynamics, chemistry, relaxation, transport, and equilibrium are evaluated pointwise, while the CFD solver keeps ownership of mesh, gradients, fluxes, time integration, and boundary conditions.

CFD-Facing Interface

The public interface is intentionally local:

`context` $\rightarrow$ `state` $\rightarrow$ `requested closures`

A reusable context is built once per mixture/model choice; a state is built per cell or quadrature point from ϱ, φ_α, and modal temperatures ϑ_ι.

Typical requests include:

- thermodynamics: p, ψ, ε_ι, mode chemical potentials,
- chemistry: species source terms τ_α,
- relaxation: modal source terms π_ι,
- transport: viscosity, conductivity, and diffusion operators,
- equilibrium: constraint-consistent local compositions via `equilibrate.py`.

The same state maps directly to conservative variables used by a solver,

$$\varrho e = \varrho \left(\sum_{\iota \in \{t,r,v,e\}} \varepsilon_\iota + \tfrac{1}{2}|\boldsymbol{v}|^2 \right),$$

with modal energy densities stored in the canonical (t, r, v, e) basis.

Mixtures, Mode Models, and Validation

Current bundled mixtures are `air5`, `air11`, and `mars19`. The same local interface can be extended to additional mixtures by supplying the corresponding thermodynamic, mechanism, and transport inputs.

Mode model	`temp_map`
1T	`[0,0,0,0]`
2T	`[0,0,1,1]`
3T	`[0,0,1,2]`
4T	`[0,1,2,3]`

Default transport closures include Chapman–Enskog viscosity and thermal conductivity with Stefan–Maxwell diffusion. The repository also carries smoke tests, regression tests, and Mutation++ comparison workflows for verification.

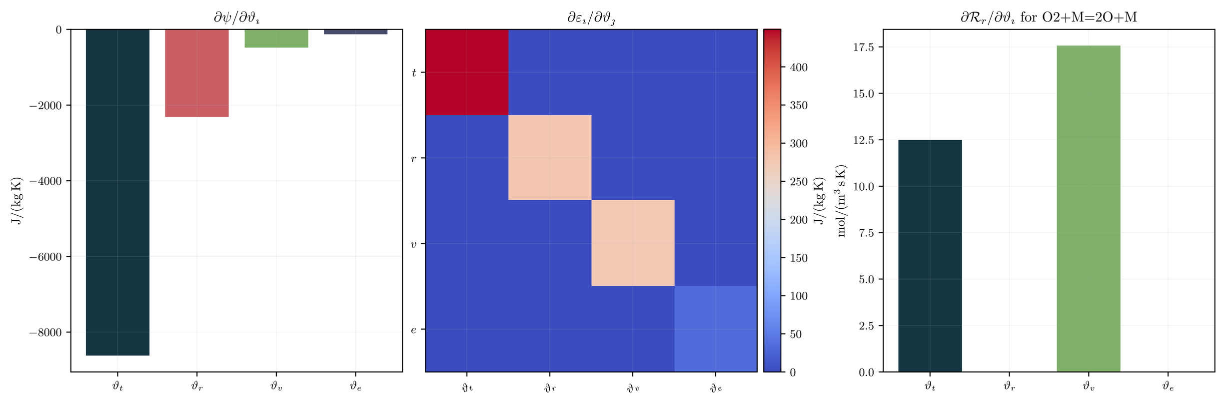

Figure 1. Selected thermodynamic and chemistry sensitivities for the baseline air5 parcel.

Conservation, Relaxation, and Equilibrium

The implementation follows a rigid-rotor harmonic-oscillator free-energy construction with modal temperatures ordered as translational, rotational, vibrational, and electronic blocks. RRHO free-energy derivatives provide the thermodynamic properties, and JAX-compatible kernels preserve batched evaluation and differentiation for derived quantities. Species production τ_α and modal exchange π_ι are exposed in conservative form:

$$\partial_t(\varrho\varphi_\alpha) + \operatorname{div}(\cdots) = \cdots + \tau_\alpha,$$
$$\partial_t(\varrho\varepsilon_\iota) + \operatorname{div}(\cdots) = \cdots + \pi_\iota.$$

Diagnostics check closed-mechanism and pure-exchange consistency through $\sum_\alpha \tau_\alpha = 0$ and $\sum_{\iota\in\mathcal{D}} \pi_\iota = 0$. The equilibrium solver in `equilibrate.py` reconstructs thermochemically consistent local compositions from the same standard-state data.

Selected Thermodynamics and Chemistry Maps

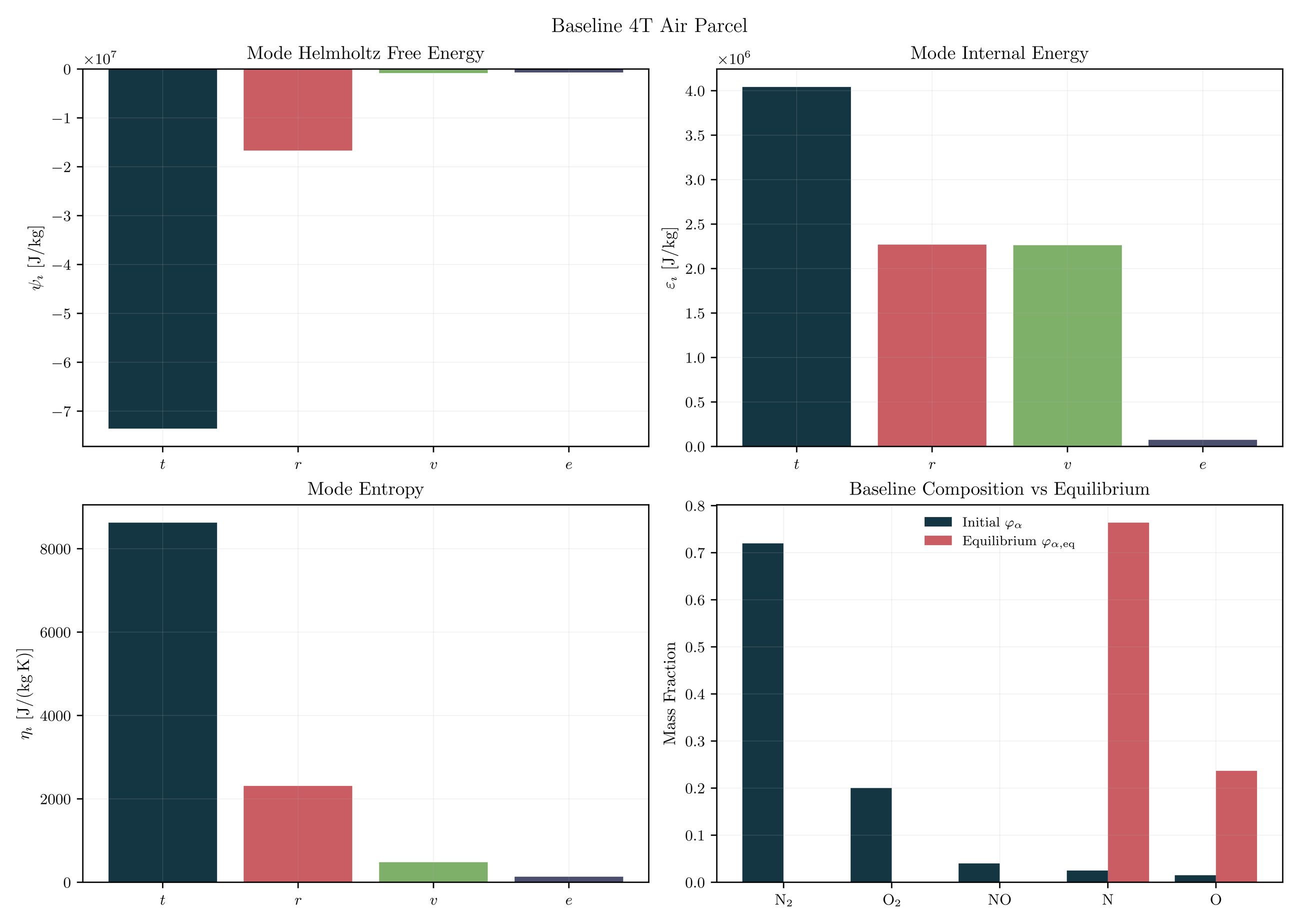

Figure 2. Baseline mode decomposition and initial-to-equilibrium composition shift for air5.

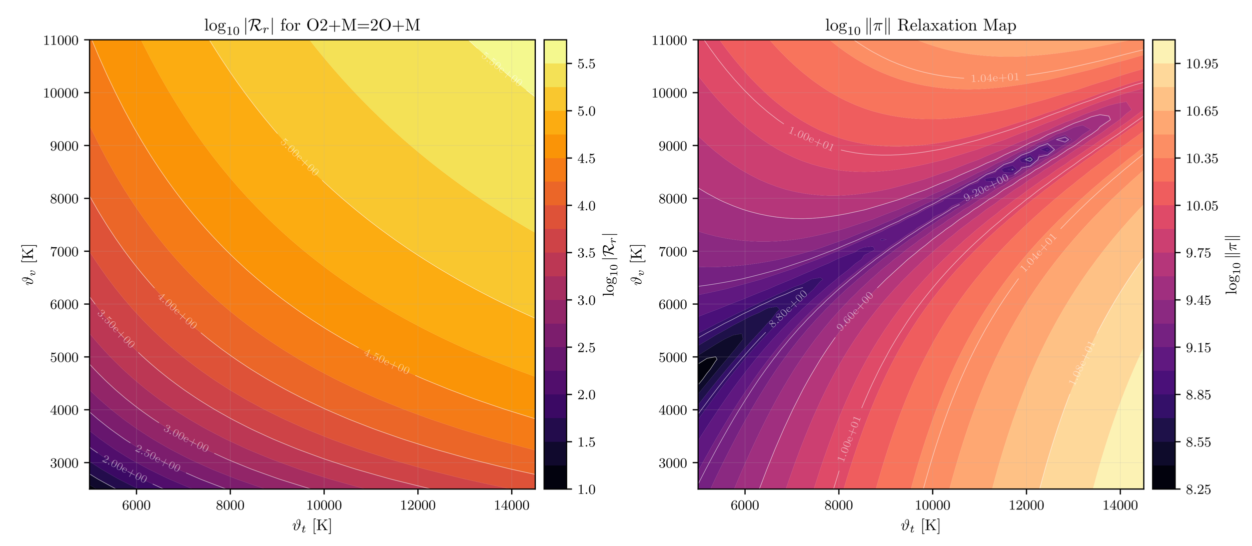

Figure 3. Finite-rate chemistry response on the same nonequilibrium plane.

Why This Separation Matters for CFD

Separating spatial numerics from stiff local physics lets a solver:

- request only the closures needed for a given model,
- detect conservation or consistency failures close to the source term,
- reuse the same local kernels across neutral air, ionising air, and Mars-entry cases.

Transport and Equilibrium Maps

Beyond scalar thermodynamic properties, the adapter returns field-shaped transport objects needed by high-temperature CFD workflows: heavy-particle conductivity, modal conductivities, multicomponent diffusion matrices, and charged-mixture diffusion variants, together with local equilibrium reconstruction.

For solver-side flux assembly, the manual exposes the local algebraic forms

$$\boldsymbol{q}_\iota = -\kappa_\iota \operatorname{grad}\vartheta_\iota, \qquad \boldsymbol{\jmath}_\alpha = -\sum_{\beta\in\mathcal{C}} M_{\alpha\beta} \sum_{\iota\in\mathcal{D}} \frac{1}{\vartheta_\iota} \operatorname{grad}\widetilde{\mu}_{\beta\iota}.$$

One mobility M serves neutral and charged mixtures. Charge adds $(\mathfrak{q}_\beta/\vartheta_{\mathrm{m}})\operatorname{grad}\phi$ to the force through $\widetilde{\mu}_{\beta\iota}$. The solver supplies $\operatorname{grad}\phi$ and assembles the flux; the library supplies M and the charge data.

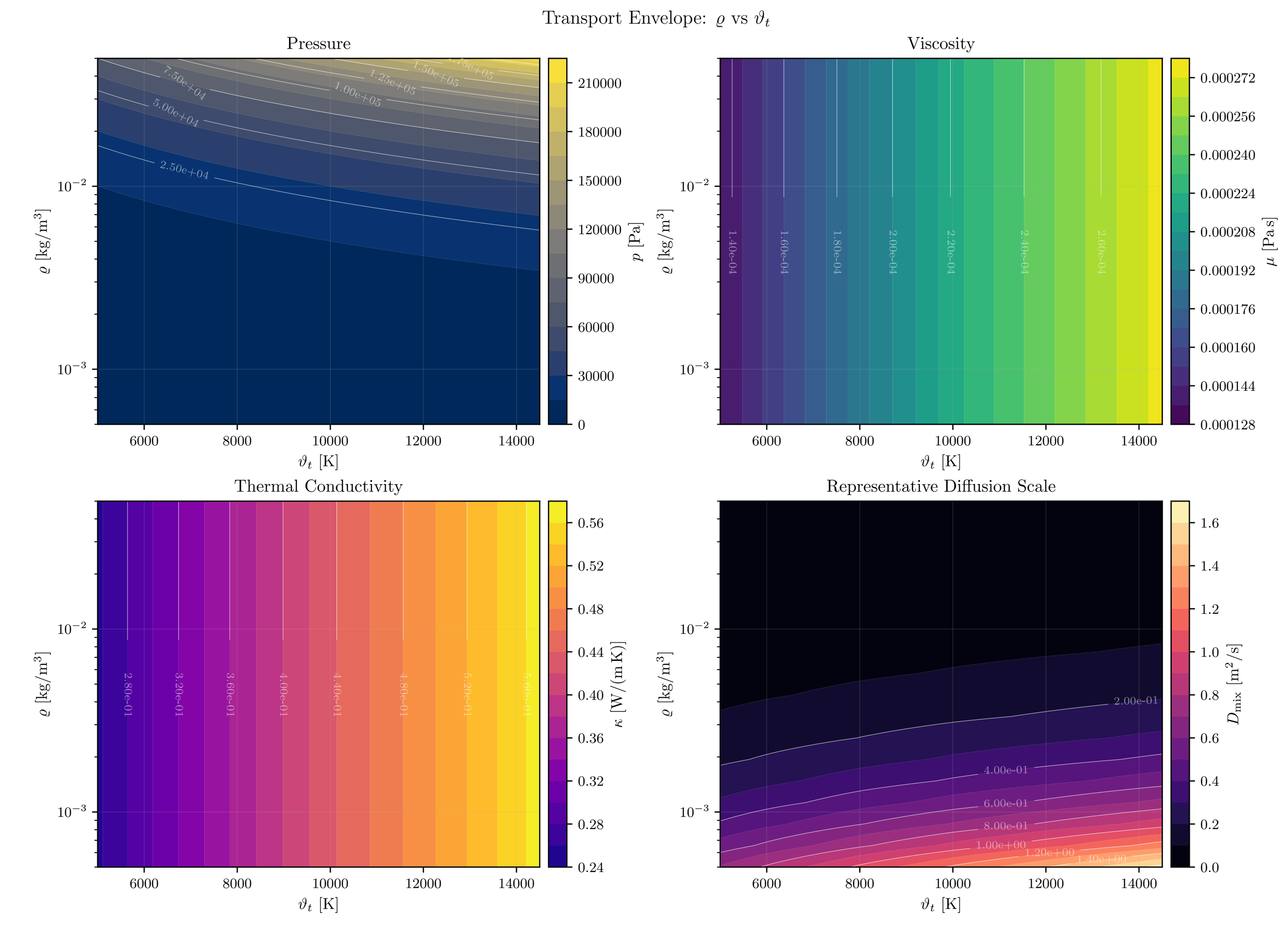

Figure 4. Air5 transport variation with density and translational temperature.

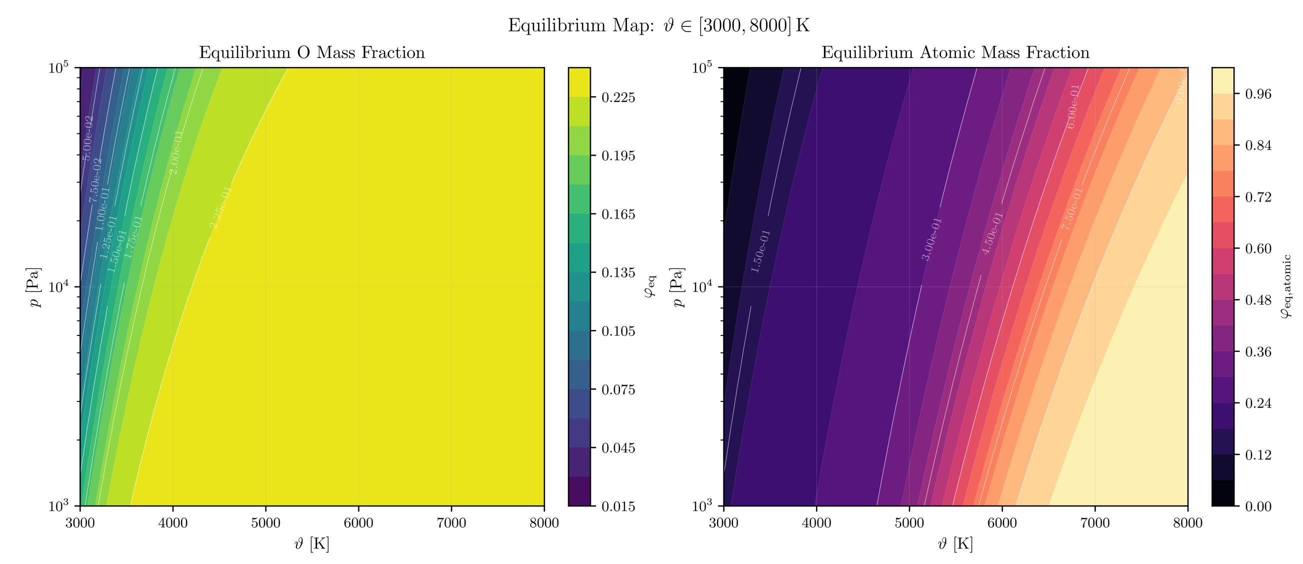

Figure 5. Focused equilibrium map for oxygen and atomic fractions over temperature and pressure.

Implementation and Verification

The library provides:

- bundled inputs and public adapters for `air5`, `air11`, and `mars19`,
- smoke tests, regression tests, and Mutation++ comparison scripts,
- consistent interfaces for thermodynamics, chemistry, transport, relaxation, and equilibrium.